# Meeting Scribe — Product Requirements Document

*A local-first, bot-free meeting transcription & document generation app for Windows*

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|---|---|
| **Document** | Product Requirements Document (PRD) v1.0 |
| **Product** | Meeting Scribe — Local Meeting Assistant |
| **Author** | Areloa |
| **Date** | May 2026 |
| **Status** | Draft — for review |
| **Target OS** | Microsoft Windows 10/11 (x64) |

# 1. Executive Summary

Meeting Scribe is a standalone desktop application for Windows that silently listens to any meeting taking place on the user's computer, transcribes it in real time, and generates polished output documents (Minutes of Meeting, action item lists, timelines, custom-template reports) — all without a bot ever joining the call.

The product is engineered around three uncompromising principles: it must run fully on-device for privacy, it must be free to operate using local open-source models, and it must travel with the user — project files are stored as portable bundles that can be opened from any device the user logs into.

### Problem

Existing meeting assistants either (a) require a bot to join the call, which is awkward, frequently blocked by IT, and visible to other attendees, or (b) ship audio to a vendor's cloud, raising privacy concerns and forcing the user onto a paid subscription once usage grows beyond a generous-looking free tier.

### Solution

A single Windows installer that captures both the user's microphone and the system's loopback audio (WASAPI), transcribes locally with Whisper, and renders documents from user-defined templates. The app is free, offline-capable, and never exposes meeting content to an external service unless the user explicitly opts in.

### Why this is different

- No bot. No meeting invite. Other participants never see the assistant.
- Fully local by default — Whisper for transcription, Ollama / Llama for document drafting.
- Free forever for personal use — no per-minute caps, no subscription gating core features.
- User-defined templates: any .docx with placeholders becomes a reusable output format.
- Portable project bundles — drop them in OneDrive / Dropbox / a USB key and resume on another machine.

# 2. Goals & Non-Goals

| Goals (in scope) | Non-goals (out of scope for v1) |
|---|---|
| • Capture mic + system audio on Windows.<br>• Transcribe locally with speaker diarization.<br>• Allow user to define output templates.<br>• Generate MoM, action items, timeline, custom docs.<br>• Persist audio + transcript + docs in a portable project bundle.<br>• Cross-device continuity via user-chosen sync folder. | • Bots that join Zoom / Teams / Meet as a participant.<br>• Real-time coaching during the meeting.<br>• Mobile (Android / iOS) — desktop only in v1.<br>• Multi-tenant SaaS or shared workspaces.<br>• CRM integrations (Salesforce, HubSpot).<br>• Live translation between languages in v1. |

## 3. Target Users & Personas

| Persona | Primary need | How Meeting Scribe helps |
|---|---|---|
| **Product / Project Manager** | Reliable minutes of meeting with action items and owners. | Custom MoM template auto-filled after each call; tasks extracted with owners. |
| **Engineer / Tech Lead** | Timeline of technical decisions made in design reviews. | Timeline view + searchable transcript stored locally with the codebase. |
| **Consultant / Solo founder** | Client-deliverable documentation without cloud exposure. | Branded .docx templates rendered locally; NDAs respected — audio never leaves device. |
| **Student / Researcher** | Free, unlimited transcription of lectures and interviews. | Unlimited local Whisper transcription with zero subscription cost. |

## 4. Competitive Landscape

The meeting-assistant market is crowded but converges on a small set of architectural choices: send audio to a vendor's cloud, transcribe there, and bill per minute. Meeting Scribe deliberately makes the opposite choice on every axis where privacy, cost or portability is in tension with vendor convenience.

| Capability | Meeting Scribe (this app) | Otter.ai | Fireflies.ai | MS Teams Live | Granola |
|---|---|---|---|---|---|
| **Bot joins meeting** | No | Yes | Yes | N/A | No |
| **Works offline** | Yes | No | No | Limited | No |
| **Audio stays on device** | Yes | No (cloud) | No (cloud) | No (cloud) | Partial |
| **Free-tier minutes/month** | Unlimited | 300 min | Limited | Requires E5 | ~25 meetings |
| **Custom .docx templates** | Yes | Limited | Limited | No | Limited |
| **Speaker diarization** | Yes (local) | Yes | Yes | Yes | Yes |
| **Cross-device continuity** | Yes (own folder) | Yes (vendor cloud) | Yes (vendor cloud) | Yes | Yes |
| **Cost per meeting at scale** | $0 | Paid above 300m | Paid | M365 license | Paid |
| **Vendor lock-in** | None — open files | Yes | Yes | Yes | Yes |

| Capability | Meeting Scribe (this app) | Otter.ai | Fireflies.ai | MS Teams Live | Granola |
|---|---|---|---|---|---|
| **Source-controllable** | Yes (.docx + .json) | No | No | No | No |

*Figures for competitors are approximate and based on publicly listed plans as of May 2026; users should verify before contractual commitments.*

### Why this matters

- Privacy: meetings often include unreleased product details, salaries, customer PII. Cloud transcription puts them in someone else's database.
- Cost predictability: bot-based tools price per minute or per seat, which scales linearly with usage. A local app has zero marginal cost.
- Behavior: bots change the meeting. People speak differently when an unfamiliar attendee labelled 'Otter' is in the call.
- Portability: vendor clouds make export friction-y. Local files are yours to move, archive, or version-control.

## 5. Functional Requirements

Requirements are tagged Must / Should / Nice using MoSCoW prioritisation.

| ID | Requirement | Priority |
|---|---|---|
| **FR-01** | Capture microphone and system loopback audio simultaneously on Windows (WASAPI). | **Must have** |
| **FR-02** | Run end-to-end fully offline; no internet connection required for core features. | **Must have** |
| **FR-03** | Transcribe captured audio locally using Whisper (faster-whisper) with selectable model size. | **Must have** |
| **FR-04** | Persist audio file, raw transcript, and metadata as a single portable project bundle (.mscribe). | **Must have** |
| **FR-05** | Allow the user to define document templates (.docx) with named placeholders that the app fills in. | **Must have** |
| **FR-06** | Generate Minutes of Meeting, action items, and a meeting timeline from any session. | **Must have** |
| **FR-07** | Speaker diarization — distinguish between speakers in the transcript. | **Must have** |
| **FR-08** | Pause and resume recording at any time during a session. | **Must have** |
| **FR-09** | Project bundles can be saved to any folder the user designates, including OneDrive / Dropbox, so the user can resume on another machine. | **Must have** |
| **FR-10** | Provide a local LLM option (Ollama) for document drafting, with an opt-in BYO- | **Should have** |

| ID | Requirement | Priority |
|---|---|---|
| | API-key path for Claude / OpenAI. | |
| **FR-11** | Detect topic shifts and tag transcript sections with a phase label (intro, discussion, decision, action). | **Should have** |
| **FR-12** | Export documents to .docx and .pdf. | **Should have** |
| **FR-13** | Full-text search across past meetings stored locally. | **Should have** |
| **FR-14** | Template marketplace / starter pack: ship with MoM, 1:1 recap, interview notes, standup, and timeline templates. | **Should have** |
| **FR-15** | Redaction of sensitive terms (regex + named-entity) before document generation. | **Nice to have** |
| **FR-16** | Audio playback synced to transcript (click a line, play that moment). | **Nice to have** |
| **FR-17** | Optional end-to-end encryption of project bundles at rest. | **Nice to have** |
| **FR-18** | Multi-language transcription (Whisper supports ~100 languages out of the box). | **Nice to have** |

## 6. Non-Functional Requirements

| Category | Target |
|---|---|
| **Performance** | Whisper-small transcription must keep pace with real-time on a 4-core x64 CPU. End-to-end document generation under 60s for a 30-minute meeting. |
| **Privacy** | Zero outbound network traffic by default. Any opt-in cloud usage must be behind an explicit toggle with a confirmation dialog. |
| **Portability** | Project bundles are self-contained ZIP files using open formats — audio (.wav/.opus), transcript (.json), generated docs (.docx/.pdf), metadata (.json). |
| **Installer footprint** | Under 500 MB installed, including the small Whisper model. Larger models downloaded on demand. |
| **Reliability** | Crash during recording must not lose audio captured so far — write to disk in rolling chunks. |
| **Accessibility** | Keyboard navigation for all core actions; high-contrast theme; screen-reader-friendly transcript view. |

## 7. System Architecture

Audio flows through seven deterministic stages from microphone to finished document. Each stage is independently testable and replaceable.

| | | |
|---|---|---|
| 1 | **CAPTURE** | Microphone via sounddevice; system audio via WASAPI loopback. Streams are mixed into a single 16 kHz mono buffer and written to disk as rolling chunks so nothing is lost on crash. |
| 2 | **SEGMENT** | Voice activity detection (silero-vad) splits the buffer into utterances. Empty / silent regions are skipped to save compute. |
| 3 | **TRANSCRIBE** | faster-whisper runs locally (small / medium / large depending on user's hardware). Output is timestamped tokens. |
| 4 | **DIARIZE** | pyannote.audio assigns a speaker label to each utterance. Labels can later be renamed by the user (e.g. Speaker 1 → 'Sarah'). |
| 5 | **STRUCTURE** | A local LLM (Ollama / Llama 3.1 8B) classifies utterances into phases (intro, discussion, decision, action) and extracts action items with owners. |
| 6 | **RENDER** | User-selected .docx template is filled in with structured data via docxtpl. Documents can be re-rendered without re-running transcription. |
| 7 | **PERSIST** | Everything (audio, transcript, structured data, generated docs) is packaged into a .mscribe bundle and saved to the user's chosen folder. |

### Local-first data flow

Stages 1–3 must run for a meeting to be useful. Stages 4–6 are deferred work that happens after the meeting ends, so they don't compete with transcription for CPU. Stage 7 is continuous — the project bundle is updated on every meaningful change.

## 8. Tech Stack

Every component listed has a free tier sufficient for unlimited personal use. There is no paid dependency on the critical path.

| Component | Purpose | Tool | Cost |
|---|---|---|---|
| **Audio capture** | Mic + system loopback on Windows | sounddevice + WASAPI | Free |
| **VAD** | Skip silence | silero-vad | Free |
| **Transcription** | Audio → text, runs locally | faster-whisper | Free |
| **Diarization** | Speaker labels | pyannote.audio | Free |
| **Local LLM** | Document drafting offline | Ollama + Llama 3.1 8B | Free |
| **Cloud LLM (opt-in)** | Higher-quality drafting | Claude API (BYO key) | BYO |
| **Template engine** | Fill user .docx templates | docxtpl + python-docx | Free |
| **PDF export** | Render docs as PDF | WeasyPrint / docx2pdf | Free |

| Component | Purpose | Tool | Cost |
|---|---|---|---|
| **Local store** | Index meetings for search | SQLite + FTS5 | Free |
| **UI** | Desktop app shell | PyQt6 (Qt for Python) | Free |
| **Packaging** | Windows installer | PyInstaller + Inno Setup | Free |
| **Sync (user-owned)** | Bundles saved to OneDrive / Dropbox | Native folder sync | Free |

*Estimated cost to operate the app indefinitely on a single device: $0.*

## 9. UI / UX Specification

The app uses a Files-style master/detail layout. The recording bar is always-on-top and intentionally minimal so the user can keep it visible during the meeting without distraction.

| Screen / Panel | Content | Behavior |
|---|---|---|
| **Home** | List of past meetings (title, date, duration, primary speaker). Big 'Start new meeting' button. | Click any row to open the meeting workspace. |
| **Recording bar** | Compact floating window: REC indicator, waveform, elapsed time, pause / stop, audio source selector. | Always-on-top. Can be moved to a second monitor. Hidden from screen-shares optionally. |
| **Meeting workspace** | Three panes: transcript (left), structured data view — action items, decisions, timeline (middle), output documents (right). | Editable transcript with speaker re-labelling. Re-generate docs without re-transcribing. |
| **Templates manager** | Browse, import, duplicate, edit .docx templates and their placeholder schema. | 'Test render' button shows what the template would produce with sample data. |
| **Settings** | Default project folder, Whisper model size, LLM backend (local / Ollama / Claude), redaction rules. | Changing the project folder migrates the index but never moves user-owned bundles automatically. |

## 10. User-Defined Template System

Templates are the keystone feature that turns Meeting Scribe from a transcription tool into a document factory. Users define what they want out of each meeting; the app fills the blanks.

### How a template is defined

1. User creates a normal Word document with their desired layout and branding.
2. Wherever they want generated content, they insert a Jinja placeholder, e.g. {{ meeting.title }}, {{ action_items }}, {{ timeline }}.
3. They drop the .docx into the Templates folder (or import via the UI).

4. The app parses placeholders, validates against the schema, and shows a 'Test render' preview.

### Built-in placeholder schema

- **meeting.title** — string, the user-set or auto-detected title.
- **meeting.date** — ISO 8601 date of the session.
- **meeting.duration** — human-readable duration.
- **attendees** — list of speaker labels (renamed by the user).
- **summary** — short executive summary written by the local LLM.
- **decisions** — list of decisions and the speaker who made them.
- **action_items** — list of {owner, description, due_date}.
- **timeline** — chronological list of {timestamp, phase, topic}.
- **transcript** — full transcript array of {speaker, start, text}.

### Shipped starter templates

- Minutes of Meeting (formal)
- 1:1 recap
- Standup digest
- Interview notes
- Decision log
- Timeline / event recap

## 11. Data Model & Cross-Device Continuity

A meeting is materialised as a .mscribe bundle — a ZIP archive whose internal layout is fully documented and stable across versions, so users (and other tools) can rely on its contents.

### Bundle layout

- **audio.opus** — compressed source audio.
- **transcript.json** — speaker-labelled timestamped utterances.
- **structured.json** — decisions, action items, timeline, summary.
- **documents/** — generated .docx and .pdf files.
- **meta.json** — title, date, duration, app version, template used.

### Cross-device continuity

The app does not run its own cloud. Instead, the user picks a default project folder that already syncs (OneDrive, Google Drive, Dropbox, or a network share). On a second machine, the user installs the app, points it at the same folder, and the local SQLite index rebuilds from the bundles found there. There is no account, no login, and no proprietary sync server — but the user can pick up exactly where they left off.

If the user starts a recording on machine A and then opens machine B, an active session lock inside the bundle prevents concurrent edits. Documents can be regenerated independently on either machine because raw audio and transcript travel with the bundle.

## 12. Risks & Mitigations

| Severity | Risk & mitigation |
|---|---|
| HARD | **Whisper performance on low-end CPUs**<br>On a 2-core machine, large Whisper models cannot keep up with real time. Mitigation: ship with whisper-small as default; only enable larger models after a hardware probe at first launch. |
| HARD | **System-audio capture permissions**<br>On Windows, WASAPI loopback works without a driver, but some pro audio interfaces (ASIO) bypass it. Mitigation: document supported interfaces; provide a fallback to capture from a virtual cable (VB-CABLE) for edge cases. |
| MEDIUM | **Local LLM quality vs. cloud LLM**<br>Llama 3.1 8B produces good but inferior summaries compared to Claude. Mitigation: BYO-key path for users who want quality, with explicit consent dialog on first use. |
| MEDIUM | **Bundle size on sync drives**<br>30-minute meetings produce ~25 MB bundles (mostly audio). 100 meetings = 2.5 GB. Mitigation: setting to store audio outside the bundle, or auto-prune audio after N days. |
| MEDIUM | **Concurrent edits across devices**<br>User opens the same bundle on two machines. Mitigation: file-lock metadata inside the bundle; warn on open if another instance has it locked. |
| LOW | **PyInstaller bundle size**<br>PyQt6 + faster-whisper + pyannote can hit ~400 MB. Mitigation: --onedir mode, ship small models by default, lazy-download larger ones. |
| LOW | **User installs without admin rights**<br>WASAPI loopback works without admin; Inno Setup supports per-user install. Mitigation: prefer per-user install by default. |

## 13. Build Plan

Seven modules, each independently testable. Don't start the next module until the current one demonstrably works.

| | |
|---|---|
| 1 | **Audio Capture** *~2 days*<br>Capture mic + system audio simultaneously on Windows using sounddevice and WASAPI loopback. Mix into a single mono stream, write to disk in rolling chunks. Goal: 10-minute test recording with no dropouts.<br>*Tools: sounddevice, pyaudiowpatch, numpy, soundfile* |
| 2 | **Local Transcription** *~3 days* |

| | |
|---|---|
| | Wire faster-whisper to the captured audio. Add silero-vad for utterance segmentation. Test small / medium / large models for accuracy vs latency. Goal: real-time transcript dumped to terminal.<br>*Tools: faster-whisper, silero-vad, ctranslate2* |
| 3 | **Diarization & Structure** *~3 days*<br>Add pyannote.audio for speaker labels. Wire Ollama with Llama 3.1 8B to extract action items, decisions, timeline from the transcript. Goal: a structured.json that captures who said what and what to do next.<br>*Tools: pyannote.audio, ollama, langchain* |
| 4 | **Template System** *~2 days*<br>Implement docxtpl-based template rendering. Define the placeholder schema. Build a Templates folder watcher. Ship 5 starter templates. Goal: select a template, click render, get a finished .docx.<br>*Tools: docxtpl, python-docx, jinja2* |
| 5 | **Project Bundles & Storage** *~2 days*<br>Define the .mscribe bundle format. Build the SQLite index for search. Implement open / save / autosave. Goal: closing the app and reopening reloads the same workspace state.<br>*Tools: zipfile, sqlite3 (FTS5)* |
| 6 | **PyQt6 UI** *~4 days*<br>Build Home, Recording bar, Meeting workspace, Templates manager, Settings screens. Wire all backend services. Goal: end-to-end UX from 'New meeting' to 'Open finished MoM in Word'.<br>*Tools: PyQt6, qt-material* |
| 7 | **Packaging & Polish** *~2 days*<br>PyInstaller --onedir build. Inno Setup installer with per-user install option. First-run hardware probe to pick Whisper model. Goal: a working .exe installer < 500 MB.<br>*Tools: PyInstaller, Inno Setup* |

### Suggested timeline

Approximately 4–5 weekends if working solo evenings and weekends. Modules 1–3 are the technically risky ones; UI work is more predictable.

## 14. Success Metrics

| Metric | Target |
|---|---|
| **Transcription accuracy** | Word Error Rate (WER) ≤ 10% on clear English audio with whisper-small. |
| **End-to-end latency** | From 'Stop recording' to 'MoM document ready' in under 60 s for a 30-minute meeting. |
| **Privacy guarantee** | Zero outbound network packets during a recorded session (verifiable via |

| Metric | Target |
|---|---|
| | Wireshark). |
| **Cost per meeting** | $0 with default local backend. Optional cloud backend itemised per session. |
| **User satisfaction** | ≥ 80% of generated MoMs require no manual edits before sharing. |

## 15. Open Questions & Future Work

- Should v2 support macOS? (Requires BlackHole virtual driver — adds a setup step.)
- Mobile companion app for browsing past meetings on the go — out of scope for v1, but the bundle format is mobile-ready.
- Marketplace for community-contributed templates (curated, signed).
- Plugin system to let other apps consume the structured.json (e.g. push action items into Linear / Jira / Notion).
- Live translation for international meetings — Whisper supports the underlying capability already.

## Appendix — Why this PRD improves on the original draft

Compared to the Sales Copilot PRD provided as input, this document strengthens the proposal in the following ways:

- Reframes the product around the user's stated needs (transcription + document generation) rather than real-time coaching, which was the original draft's focus.
- Replaces the cloud transcription dependency (Deepgram) with a local Whisper pipeline so the app is genuinely standalone and free.
- Adds a first-class user-defined template system with a documented placeholder schema.
- Defines a portable .mscribe bundle format so the user can move between devices without a proprietary cloud.
- Adds a full competitive landscape table benchmarking against Otter, Fireflies, Teams Live, and Granola.
- Expands risks beyond packaging to cover the harder issues: local model performance, audio capture edge cases, and concurrent device usage.
- Explicit personas, goals/non-goals, and success metrics — making scope easier to defend during build.